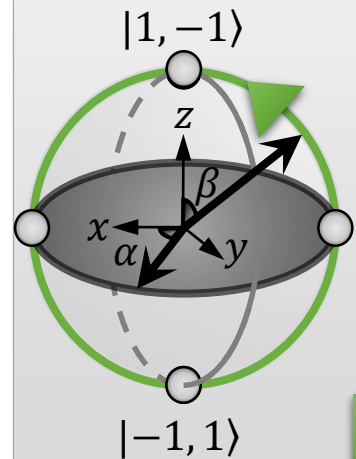
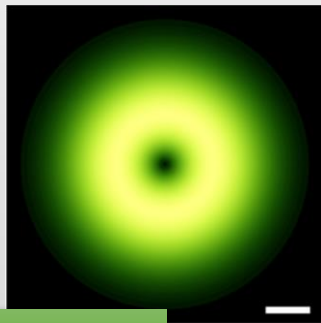


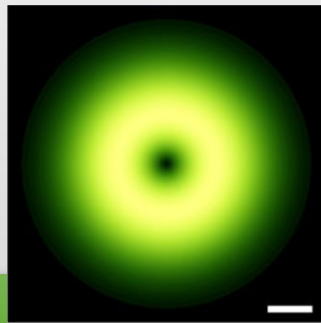
# Valley-mixed excitons transition rate $\Gamma_{B\pm, \mathbf{Q}}^{1,-1,-1,1}(\alpha, \beta)$



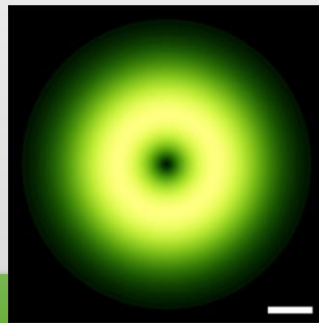
$\beta = 0$



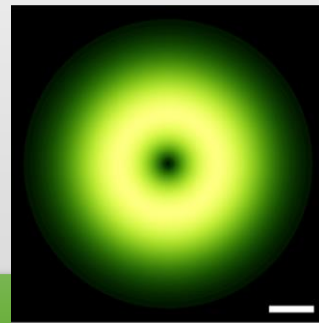
$\beta = \pi/2$   
 $\alpha = 0$



$\beta = \pi$



$\beta = \pi/2$   
 $\alpha = \pi$



meridian